



Wrap Curve

Wrap flat curves onto a curved profile by arc length -- length along the profile kept, height and width carried over.

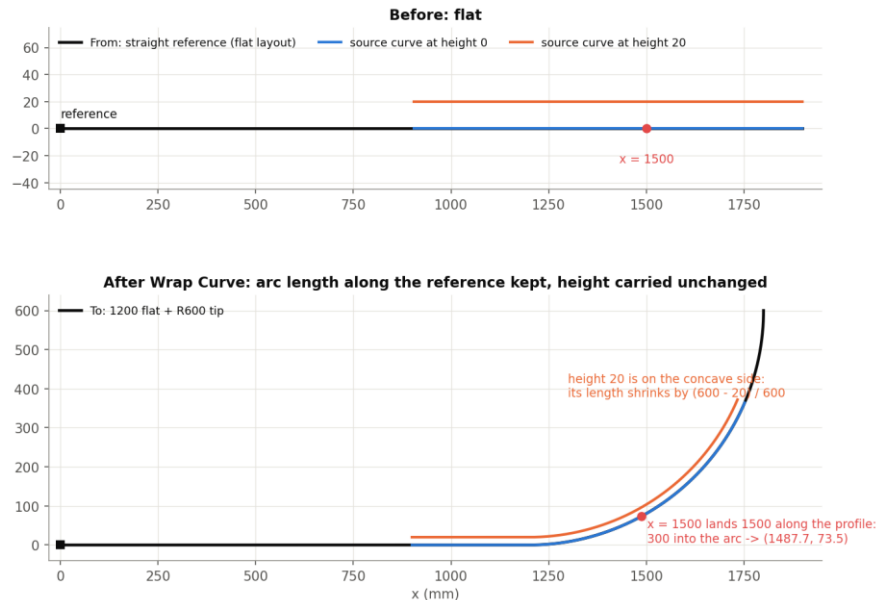


What it does

Planforms, graphics, inlays and insert lines are drawn flat; the ski is built on a rockered, cambered profile. Native Wrap only handles cylinders and cones, and projecting foreshortens. Wrap Curve keeps the arc length along the profile, so a station 1500 mm along the layout lands 1500 mm along the ski.

Use it when

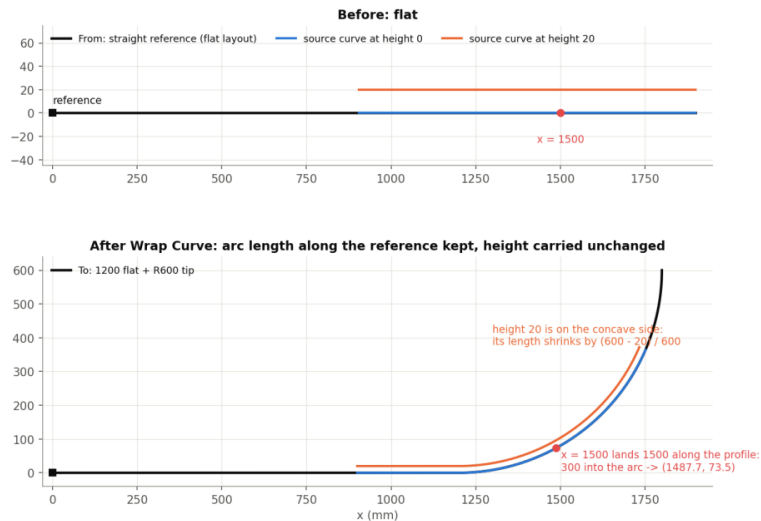
- Put flat layout lines on the profile
- Keep stations at their true distance along the ski
- Carry width and height exactly



Tip example: 1200 flat + R600 tip; x = 1500 lands 300 into the arc.



The bending map



Arc length kept

A point s along the From reference lands s along the To reference (a shift, no scaling).

Height and width kept

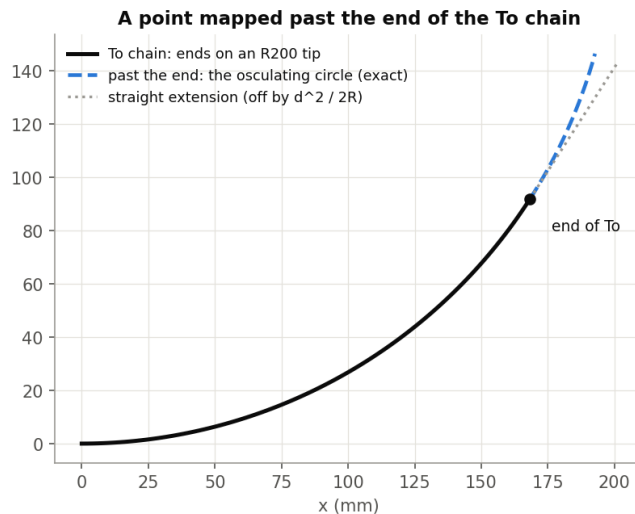
The point's offsets in the frame carry over: ribs stay perpendicular.

Layers bend

At height h around radius R they stretch or shrink by $(R - h) / R$, as a real layer does.



Frames and ends



Frenet (default)

One normal carried along with minimal rotation: no flip at inflections.

Binormal

Normal from a plane you give (face or mate connector); every edge must lie in it.

Past the ends

The osculating circle continues the chain -
- exact on arcs.



The dialog

IMAGE MISSING:
docs/decks/wrap_curve/shots/dialog.png

#	Parameter	What it means
1	From data	From edge(s) (G1 chain) and From reference (vertex, mate connector, plane).
2	To data	To edge(s), Flip (reverse To), To reference, Flip normal.
3	Source curves	The curves to wrap.
4	Frame orientation	Frenet (default) / Binormal (face, mate connector or world axis).
5	Details	Sampling (control-point or length based), degree 3, max control points 15, tolerance 0.01 mm; Break spans at curvature jumps and corners.
6	Debug	Print curves, show frames and sample points.



Outputs and messages

Outputs

Key	What it holds
(wire)	One wire: spline spans, plus exact copies on straight stretches.

Messages

	Message / what to do
Console	span fitted at a lower degree The source's degree was too high for a short To edge.
Console	cannot be fitted; the output will have a gap Rebuild the source at degree 3 or sample by length.



Tips and limits



Flip normal

If 'up' comes out on the wrong side -- older models may need it after the 2026-09-23 frame change.



Only To flips

From's direction comes from its chain; an off-centre From reference pushes points past an end.



Break spans

Changes topology (more edges) -- downstream references may move.



Where it lives

Document	curveMapping_public (d12583277c42b00c63a9b7b7)
Tab	wrapCurve (+ engine curveMappingCore, pinned)
Tests	none
Explainer	docs/explainers/curve_mapping/curve_mapping_explained.md, section 1, 2.1
Related	Wrap Curve, Wrap and Loft, Deform, Measure curve distance, Offset edges